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Advanced Quantitative Methods

Predicting Post-Graduation Earnings at U.S. Four-Year Institutions

Research Question

In looking at education institutional outcomes, we are interested in investigating characteristics associated with stronger labor-market outcomes for graduates. This study investigates the following research question: What institutional characteristics predict median earnings 10 years after entry at four-year colleges in the United States?

Several factors may influence long-term earnings outcomes. An area of particular interest is whether small private institutions' selectivity shapes the academic ability of admitted students, which can influence future labor market success of graduates. The socioeconomic composition of students may also be associated with differences in career networks and resources. Institutional scale and sector (public vs. private) may further influence resources, educational quality, and job placement opportunities.

Based on these considerations, this analysis examines how institutional selectivity, student socioeconomic characteristics, and institutional size relate to median earnings ten years after students first enroll.

Variables

Six independent variables were selected based on theoretical relevance to post-graduation earnings.

1. Percent Pell Grant recipients (pctpell)
   This variable represents the proportion of students from lower-income households. Institutions serving larger shares of low-income students may have graduates with lower average earnings due to differences in socioeconomic opportunity structures. A negative relationship with earnings is expected.
2. Pell category (pell_cat)
   This categorical variable groups institutions by low, medium, or high Pell participation. Institutions with higher Pell concentrations may have lower average earnings outcomes. A negative relationship is expected for higher Pell categories.
3. Percent receiving federal loans (pctfloan)
   Higher loan participation may indicate greater financial need among students. Institutions with more students borrowing heavily may see lower post-graduation earnings. A negative relationship is expected.
4. Admission rate (adm_rate)
   Admission rate reflects institutional selectivity. More selective institutions (lower admission rates) often enroll students with stronger academic preparation and may provide stronger career networks. Therefore, admission rate is expected to have a negative relationship with earnings.
5. Enrollment size (enrollment)
   Smaller institutions may be more selective in who is admitted into their program and

have higher enrollment barriers for low-income prospective students. As a result, a negative relationship between enrollment and earnings is expected.

6. Institutional type (is_public)
   Public and private institutions often differ in funding structures, selectivity, and resources. Private institutions may provide stronger labor market outcomes on average, suggesting a negative coefficient for public institutions when compared with private ones.

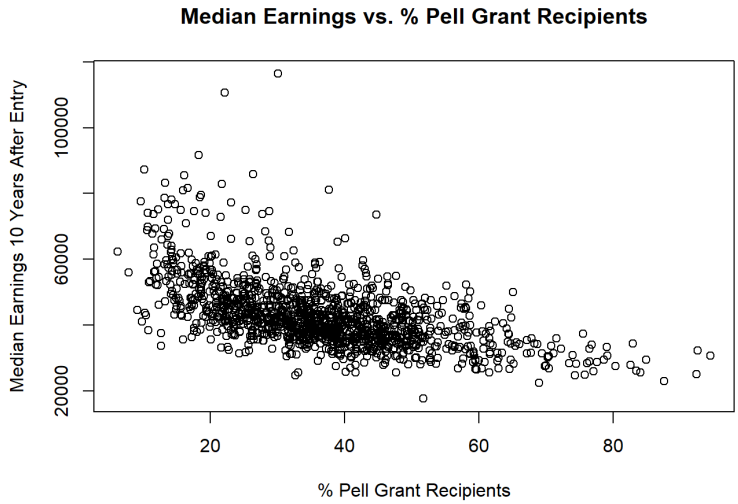
Data Cleaning and Initial Checks

Basic descriptive statistics and scatterplots were used to evaluate the data. Summary statistics indicated that the dataset contains 1,326 institutions, with admission rates ranging from roughly 6% to 100% and enrollment ranging from approximately 126 to over 50,000 students. Median earnings ten years after entry vary widely across institutions.

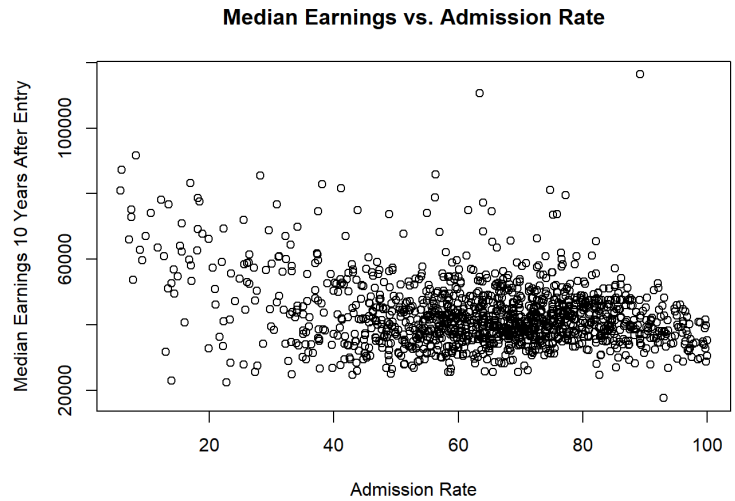
The dataset did not contain problematic out-of-range values for the variables examined. Any missing observations were removed using listwise deletion to ensure consistency across regression models. A check of institution identifiers confirmed that there were no duplicate institutions.

Scatterplots comparing earn10yr with each candidate variable revealed several patterns:

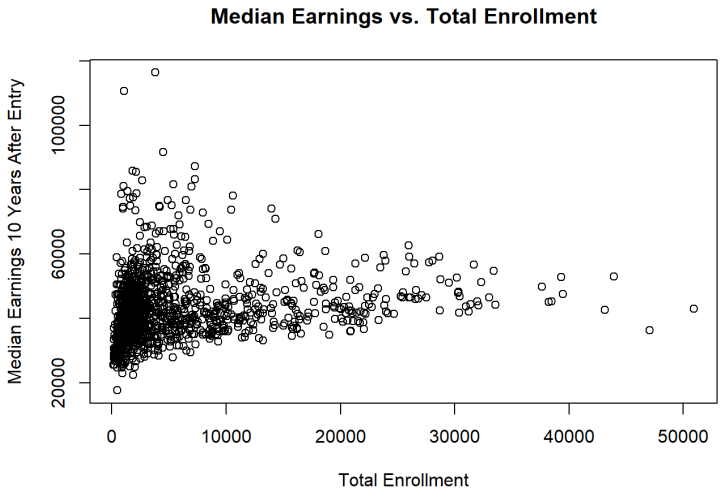
1. Institutions with higher percentages of Pell recipients tended to have lower earnings outcomes.



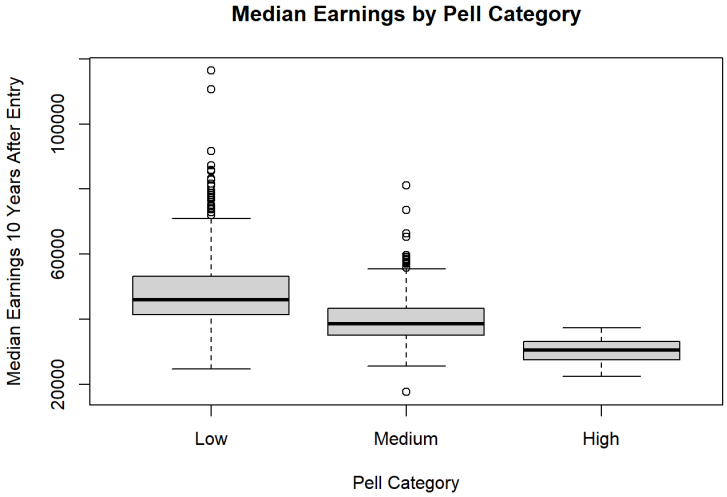
2. Admission rate displayed a negative relationship with earnings, indicating that more selective institutions tend to produce higher earnings.



3. Enrollment size showed a weak positive relationship with earnings.



4. A noticeable difference between median earnings can be observed by Pell Grant Category boxplots.

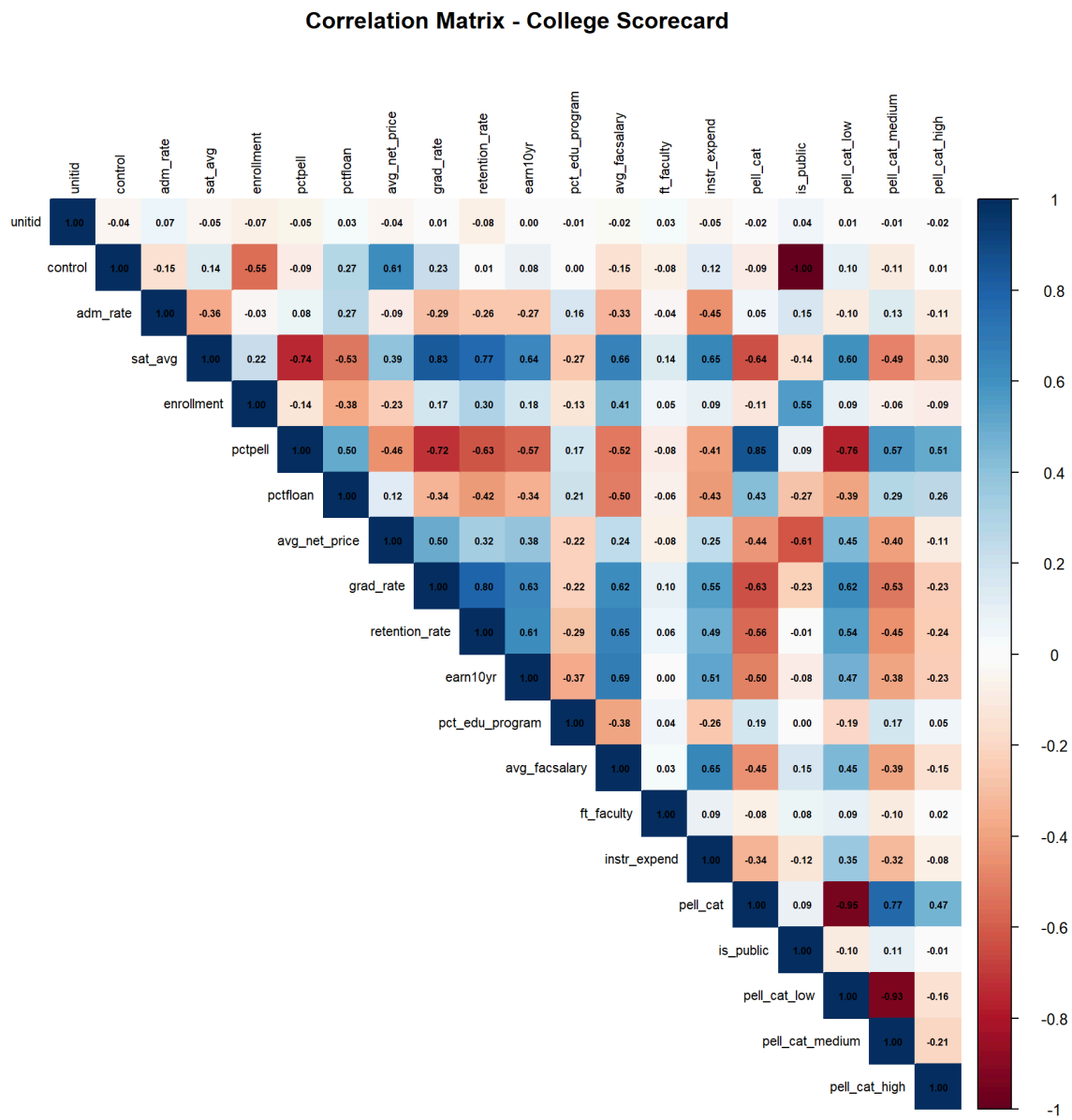


These exploratory visualizations suggested that socioeconomic composition and selectivity may be particularly important predictors of earnings.

Exploratory Data Analysis

A correlation matrix was used to examine relationships among the variables.
The strongest correlations with earn10yr were observed for:

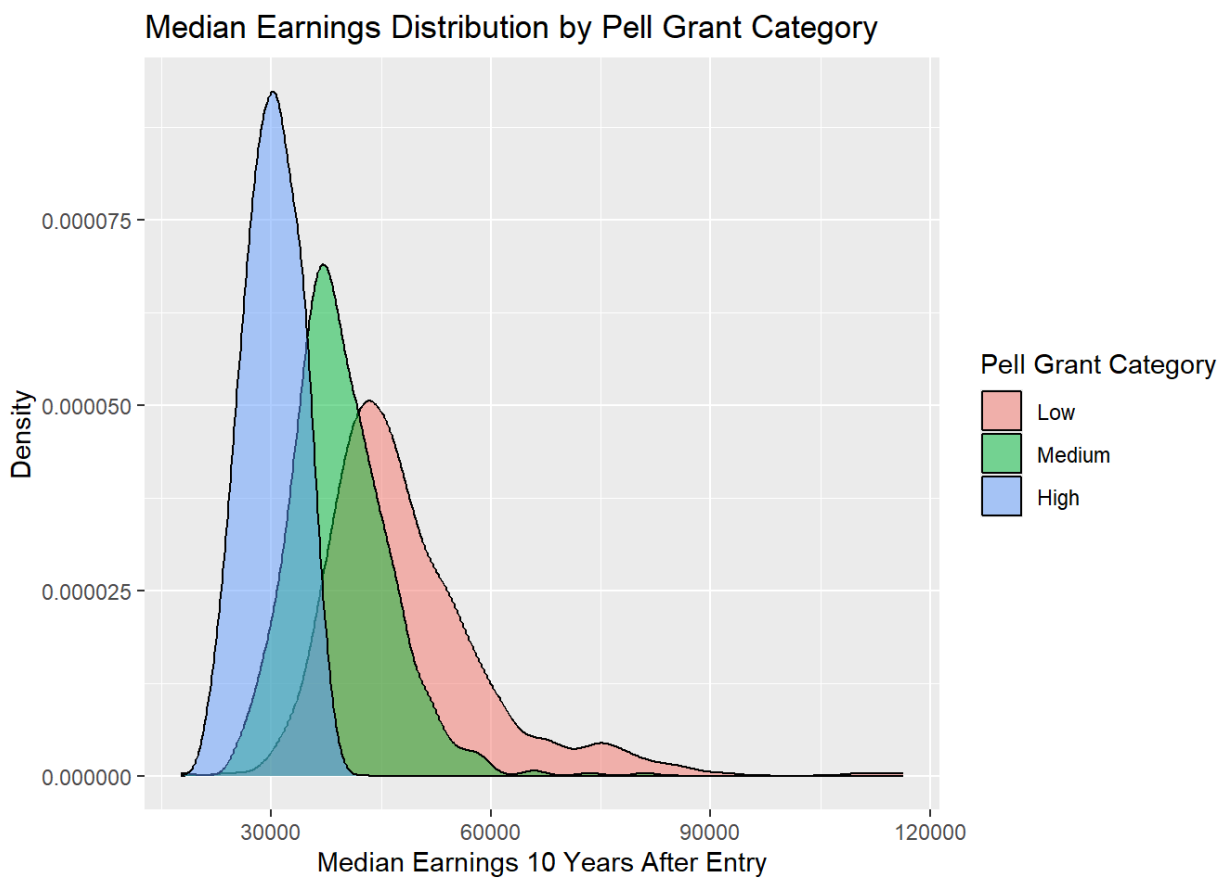
- Percent Pell recipients, which showed a strong negative correlation with earnings.
- Admission rate, which also showed a negative relationship with earnings.
- Enrollment, which showed a smaller positive relationship.



These findings were largely consistent with expectations. One notable observation was that the correlation between percent federal loan recipients and earnings was relatively weak, suggesting that loan participation may not strongly predict earnings outcomes.

Correlations among independent variables were examined to assess potential multicollinearity. Although some relationships existed between socioeconomic indicators, none were high enough to immediately suggest severe multicollinearity concerns.

A density plot of earnings by Pell category revealed a clear distributional pattern.



Institutions with low Pell participation generally had higher earnings distributions, while institutions with high Pell participation were more concentrated at lower earnings levels. This supports the hypothesis that socioeconomic composition is strongly related to long-term earnings outcomes.

Final Model Plan

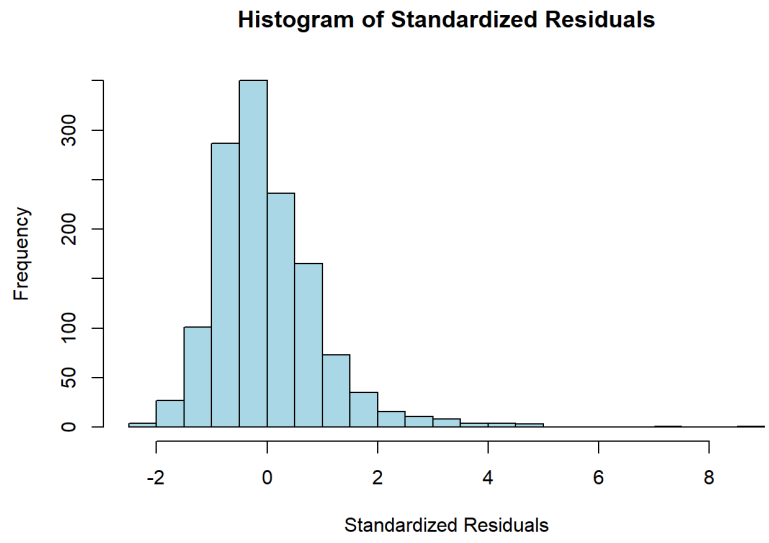
Based on the exploratory analysis, the original set of independent variables was retained. The ordering was also preserved, as socioeconomic variables appeared to be the strongest predictors and were therefore appropriate to enter first.

Bivariate Regression

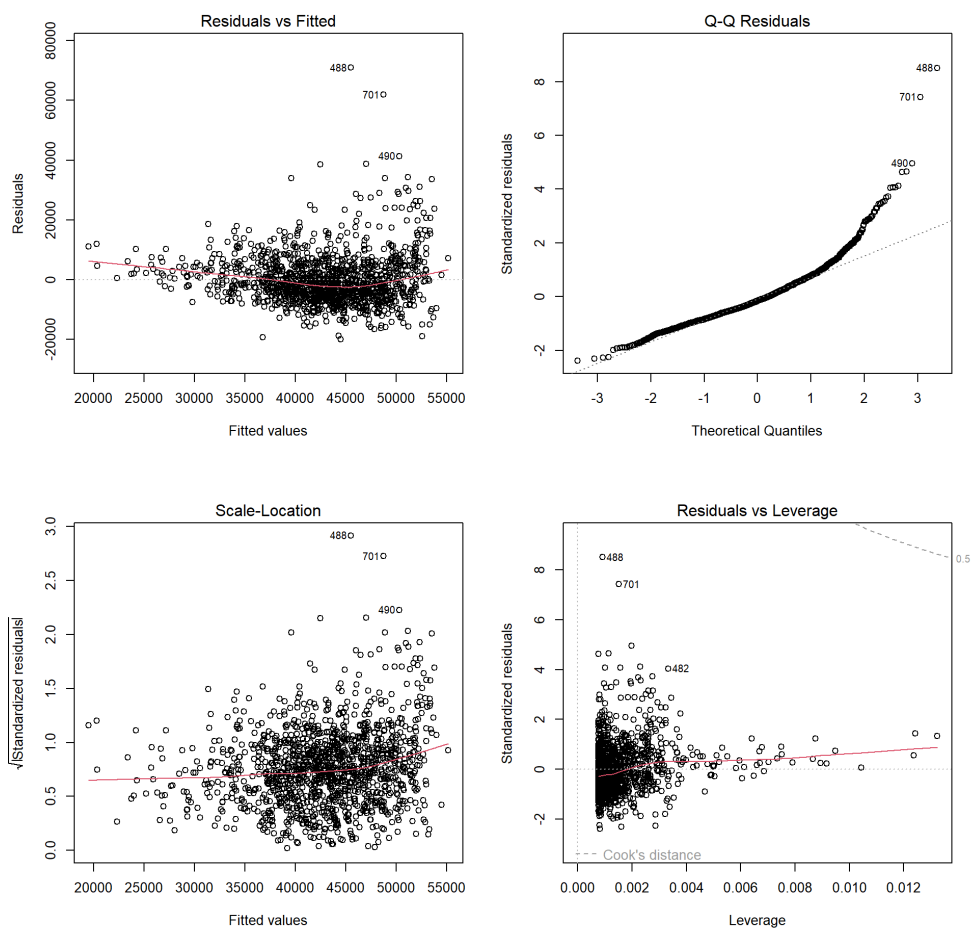
The first regression model examined the relationship between percent Pell recipients (pctpell) and median earnings 10 years after entry. The coefficient for pctpell was negative and statistically significant, indicating that institutions with higher percentages of Pell recipients tend to have lower median earnings outcomes. The p-value was well below conventional significance thresholds, indicating a statistically significant relationship between the variables. The R-squared value indicated that a meaningful portion of the variance in earnings could be explained by this single predictor. This supports the idea that student socioeconomic composition plays a substantial role in long-term earnings outcomes. The direction of the relationship matched the original theoretical expectation.

Bivariate Diagnostics

Several diagnostic tests were conducted to evaluate the regression assumptions. The histogram of residuals suggested that the residuals were approximately normally distributed, although some skewness was present due to a few institutions with exceptionally high earnings outcomes.



The residuals versus fitted values plot did not show severe heteroscedasticity, although some widening of variance at higher earnings levels was visible.



Cook's distance indicated a small number of potentially influential observations, but none were large enough to dominate the model. The Durbin-Watson test did not suggest serious autocorrelation in the residuals, which is expected because institutions represent independent observations. A small number of standardized residuals exceeded ±2, indicating some outliers but not enough to invalidate the model. Overall, the diagnostics suggested no major violations of regression assumptions in the bivariate model.

Final Model

The final regression model included:

1. Percent Pell recipients
2. Pell category indicators
3. Percent federal loan recipients
4. Admission rate
5. Enrollment size
6. Public vs private status

In the final model:

- **Percent of students receiving Pell Grants** (pctpell) is a strong negative predictor ($\beta = -349.4$, $p < .001$). Colleges with higher proportions of Pell recipients tend to have lower average post-graduation earnings.
- **Admission rate** (adm_rate) is also negatively associated with earnings ($\beta = -114.98$, $p < .001$), suggesting that more selective institutions (lower admission rates) are associated with higher graduate earnings.
- **Enrollment size** (enrollment) shows a small but significant positive effect ($\beta = 0.209$, $p < .001$), meaning larger institutions tend to have slightly higher average earnings outcomes.
- **Institution type** (is_public) shows a noticeable difference, with public institutions showing significantly lower predicted earnings compared with private institutions ($\beta = -1584.44$, $p = .0069$).

The final regression model was statistically significant ($p < 0.001$) with about 39% (Adjusted R-squared = 0.386) indicating that the model explains about 39% of the variation in post-graduation earnings across institutions. Other predictors, such as Pell category indicators and percent federal loan recipients, were not statistically significant in the final model.

Final Model Diagnostics

Diagnostic tests were repeated for the full model. The residual distribution remained approximately normal, although some skewness persisted due to institutions with extremely high earnings outcomes. The residuals versus fitted plot did not reveal severe heteroscedasticity. Only a modest number of standardized residuals exceeded ±2, suggesting that outliers were present but not excessively influential.

Variance Inflation Factor (VIF) values were examined to assess multicollinearity. All values were well below common thresholds (5 or 10), indicating that multicollinearity is not a major concern.

The Durbin-Watson statistic indicated that the assumption of independence of errors was satisfied. Taken together, these diagnostics suggest that the model is statistically sound and reasonably trustworthy, though a substantial portion of variation in earnings remains unexplained.

Interpretation

Reflecting on my initial research question from Section 2, most of my expectations were supported by the final regression results. I anticipated that a higher percentage of Pell grant recipients (pctpell) at an institution would be linked to lower graduate earnings, and the analysis confirmed this association, likely reflecting broader socioeconomic challenges faced by these students. I did not anticipate total enrollment to have a positive relationship with median earnings 10 years after entry. The coefficient of 0.2516 suggests that for every 1000 enrolled students, the predicted median earnings are expected to increase by $2516. This runs contradictory to my hypothesis that small private schools would be associated with higher median earnings. This could be due to greater resources, opportunities, or alumni networks at larger institutions, which the results help reflect.

Additionally, my hypothesis that more selective institutions (with lower admission rates) would have graduates with higher earnings held true, aligning with the idea that selectivity is associated with other valuable institutional qualities. What surprised me, however, was the relative strength of these relationships, especially how pronounced the association was for selectivity and enrollment. I also found it notable that after controlling for factors, these key predictors (pctpell, adm_rate, enrollment, and is_public) remained the most influential.

Limitations

Although this regression analysis offers valuable findings, it is important to recognize several key limitations:

1. Correlation Does Not Imply Causation: Because the analysis relies on observational rather than experimental data, any associations identified cannot be taken as evidence of causality. The relationships observed between institutional characteristics and graduate earnings may be influenced by underlying factors not included in the model.
2. Potential for Omitted Variables: There may be important factors affecting post-graduation earnings—such as academic rigor, local job markets, or student demographics—that were not measured or included. Excluding these variables could bias the estimated effects of the predictors used.
3. Limits to Generalizability: The model is built on a specific dataset covering certain colleges and a particular cohort (10-year earnings). As such, its findings may not extend to all higher education contexts, different cohorts, or international settings.
4. Multicollinearity and Specification: Even though VIFs suggest multicollinearity is not severe, some predictors may still be correlated, complicating interpretation of individual effects.
5. Outliers and Model Robustness: While outliers were identified based on standardized residuals, they were not specifically adjusted for or examined further, which could impact the reliability of the results.

Conclusion

This analysis examined which institutional characteristics predict median earnings ten years after entry at four-year colleges. The results indicate that student socioeconomic composition and institutional selectivity are among the strongest predictors of earnings outcomes.

Institutions with higher percentages of Pell Grant recipients tend to have lower median earnings outcomes, while more selective institutions tend to produce higher earnings. Institutional size also shows a modest positive relationship with earnings, and public institutions appear to have slightly lower earnings outcomes compared with private institutions.

Although the model explains roughly 39% of the variation in earnings, many other factors (such as academic major, regional labor markets, and institutional resources) likely influence graduate earnings as well.

Overall, the findings highlight how both institutional characteristics and student socioeconomic composition are strongly associated with long-term labor market outcomes. Further research with more comprehensive data, carefully selected variables, and alternative modeling approaches is needed to more fully understand the factors influencing graduate earnings.